

## ORIGINAL ARTICLE OPEN ACCESS

# Leveraging Large Language Models to Enhance Self-Regulated Learning in Programming Education With Explainable AI

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## ABSTRACT

**Background:** While prior research has shown that timely and personalised feedback improves students' learning outcomes and self-regulation, most existing systems fail to provide actionable, individualised explanations at scale, especially in programming education. Manual feedback is resource-intensive, and traditional Artificial Intelligence (AI) systems often lack transparency, limiting their pedagogical value.

**Objectives:** This study addresses these gaps by leveraging Large Language Models (LLMs) and Explainable AI (XAI)—specifically, the SHapley Additive exPlanations (SHAP) method—to generate interpretable, scalable feedback that enhances self-regulated learning (SRL) in the context of programming education.

**Methods:** In the present study, behavioural data were collected from BookRoll, an e-reading system that tracks interactions like highlighting and note-taking, and VisCode, a coding platform that records compile attempts, error types and code execution behaviour. Combined with self-reported strategy data, these formed the LBLS dataset used to train a predictive model. SHAP was used to identify key learning features, which were then input into the GPT-4 model to generate personalised mid-semester reports.

**Results:** Results showed that students receiving LLM-generated suggestions improved in SRL behaviours and final performance. Most found the feedback understandable and useful, though some questioned its accuracy.

**Conclusion:** This study demonstrates the potential of combining LLMs and XAI to deliver meaningful, scalable feedback, but also highlights the need for human oversight.

## 1 | Introduction

Explainable Artificial Intelligence (XAI) has gained considerable attention in educational research for its ability to enhance transparency and trust in AI-driven predictions. By offering interpretable outputs, XAI helps educators and learners understand the reasoning behind predictions and supports informed instructional decisions (Türkmen 2024). In the context of learning analytics,

XAI is defined as a set of techniques that makes the decision-making pathways of AI systems transparent and understandable to human stakeholders (Li et al. 2024). Specifically, it reveals the feature contributions that drive a model's prediction, preventing the 'black box' problem from hindering pedagogical trust. Still, many studies rely on post hoc explanations that do not integrate into students' learning processes or evaluate their pedagogical impact (Hooshyar and Yang 2024; Jeyasothy et al. 2023).

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## Practitioner Notes

- What is currently known about this topic?
  - Self-regulated learning (SRL) is crucial for student success in programming education.
  - Timely and personalised feedback is key to supporting SRL, but traditional manual feedback is difficult to scale.
  - While AI can be used for feedback, its 'black box' nature often makes it difficult to interpret, reducing its educational value.
  - Explainable AI (XAI) can increase the transparency of AI predictions, but its outputs can be too complex for non-technical learners.
- What does this paper add?
  - This study presents an innovative approach that combines the analytical power of XAI (SHAP) with the natural language generation capabilities of a Large Language Model (GPT-4) to provide scalable and comprehensible personalised feedback in programming education.
  - The research confirms that this combined XAI-LLM method can effectively translate complex learning behaviour data into specific, actionable suggestions that students can understand and adopt.
  - Findings indicate that students who received LLM-generated feedback showed significant improvements in self-regulated learning strategies (such as help-seeking, rehearsal and metacognitive monitoring) and academic performance in programming.
- Implications of study findings for practitioners
  - Educators and instructional designers can consider adopting systems that integrate XAI and LLMs to provide students with large-scale, automated and personalised learning support.
  - In programming courses, providing feedback that explains how student learning behaviours impact their performance can effectively promote metacognition and self-regulation.
  - While AI-generated feedback has great potential, educators still play a critical role in the system by supervising the quality and accuracy of the feedback and providing additional guidance when necessary.
  - When developing or selecting AI educational tools, priority should be given to systems that not only provide predictions but also explain the reasoning behind them in a learner-centric manner.

Simultaneously, Large Language Models (LLMs) have emerged as impactful tools across education due to their capacity to generate natural language explanations and personalised feedback. Research suggests that integrating LLMs with XAI bridges the interpretability gap, offering learner-centred insights from complex models (Steinert et al. 2024). This synergy supports adaptive, formative feedback aligned with students' individual needs (Pozdniakov et al. 2024).

These developments are particularly relevant in Self-Regulated Learning (SRL), which is a critical student competence that underpins success in programming education. Programming requires students to plan, monitor and adjust strategies—core SRL behaviours supported by timely, personalised feedback (Sun et al. 2025). LLMs have shown promise in fostering SRL through scaffolded, reflective feedback (Wong and Viberg 2024), yet few studies compare the interpretability or learning impact of LLM-based versus XAI-based explanations. It is crucial to distinguish this approach from commercial AI coding assistants like GitHub Copilot or Amazon CodeWhisperer. While these 'copilots' provide tactical, real-time code completion to accelerate implementation, they often bypass the struggle required for deep learning. In contrast, our system is designed to provide strategic, metacognitive support—helping students reflect on how they are learning (e.g., their debugging strategies) rather than simply giving them the code.

To address this gap, the present study first trained a predictive model using the LBSL-516 dataset, a pre-existing dataset containing learning behaviour and strategy data from 516 prior students. This trained model was then used in our main experiment, where we integrated XAI with LLMs to produce and evaluate individualised learning reports for a new cohort of students. The learning behaviour data were collected from two online platforms: BookRoll, which logs students' reading interactions with e-materials, such as highlighting and note-taking (Ogata et al. 2015); VisCode, which tracks programming activities, including code compilation, copy-paste actions and error occurrences in real time (Lu et al. 2016). The learning strategy data were collected using two instruments. (1) The Motivated Strategies for Learning Questionnaire (MSLQ) assesses students' motivation and learning strategies across 15 dimensions, including goals, self-efficacy, anxiety, metacognition and resource management (Pintrich et al. 1991). (2) The Strategy Inventory of Language Learning (SILL) evaluates six categories of language learning strategies, such as cognitive, metacognitive and social strategies (Oxford 1991).

In this framework, XAI is first used to extract interpretable features from predictive models (such as Random Forest and Support Vector Machines), and these are then processed by an LLM to generate accessible feedback and improvement suggestions. The System Causability Scale (SCS)—a measure designed to assess how well users can understand and trust the causal reasoning in AI-generated feedback—is used to evaluate the comprehensibility of these explanations, and their effect is measured through changes in SRL strategies and programming performance. This study therefore aims to answer the following research questions:

**RQ1.** Does the integration of LLMs with XAI outputs provide superior interpretability compared to standalone XAI visualisations in programming education?

**RQ2.** Can LLM-generated improvement suggestions enhance students' self-regulation and programming language learning strategies?

**RQ3.** Can LLM-generated improvement suggestions improve students' learning performance in programming education?

## 2 | Literature Review

### 2.1 | XAI in Education and Its Benefits for Learners

As AI technologies become more prevalent in education, one of the major concerns has been the lack of transparency behind how AI models make decisions, which often leads to low trust and hesitation from both teachers and students. XAI addresses this by offering insights into why and how predictions are made—allowing users to better understand system logic and thereby increasing acceptance and ethical alignment in educational settings (Li et al. 2024). XAI helps convert opaque machine outputs into accessible explanations, which can be critical when applied to learner modelling, feedback systems, or early-warning interventions for at-risk students (Berendt et al. 2020; Rawal et al. 2023). In other words, learners don't just get a score—they get a sense of how their behaviour led to that outcome, and that actually changes how they study or prepare.

The benefits of XAI go beyond technical interpretability; they also support learner empowerment and metacognition. By providing tailored explanations of learning outcomes, students can better reflect on their strengths and weaknesses, adjust strategies and take ownership of their progress. Recent research has shown that usable XAI approaches—when designed with learner comprehension in mind—can significantly enhance student motivation and academic self-efficacy (Harry 2023). In practical terms, this could mean showing a student not just that they underperformed, but which learning behaviours—such as skipped readings or low practice time—contributed to the result. Moreover, XAI plays a key role in ensuring fairness and accountability when AI is deployed at scale in schools, especially when integrated into feedback dashboards or intelligent tutoring systems (Lohr et al. 2025; Surugiu et al. 2024). However, XAI systems still face issues in being overly technical or abstract, and efforts must be made to ensure explanations are both accurate and understandable for diverse learners (Barman et al. 2024; Kay 2012).

### 2.2 | The Integration of LLMs With XAI

The integration of LLMs and XAI represents a powerful advancement in educational technology, particularly in making complex AI reasoning processes more accessible to learners. While traditional XAI methods such as SHAP or LIME explain model behaviour in terms of statistical contributions, they often fall short in user comprehension due to abstract visualisation or technical jargon. As Hooshyar and Yang (2024) pointed out, despite the popularity of SHAP and LIME in educational settings, their reliance on statistical visualisations and post hoc logic can mislead users or obscure model reasoning when used without sufficient contextualisation. This highlights the need for alternative approaches that improve both interpretability and causal transparency in feedback delivery. LLMs, by contrast, can translate these explanations into natural language narratives that are easier to understand and act upon. This synergy enables XAI to reveal important features contributing to predictions, and LLMs to interpret and contextualise these findings for human users (Lohr et al. 2025; Steinert et al. 2024).

Recent research underscores the pedagogical advantages of this integration. For instance, Hsu et al. (2024) proposed a method for translating SHAP visualisations into textual explanations generated by LLMs, aiming to enhance model transparency and improve comprehension for non-technical users. Similarly, Koutchme et al. (2025) demonstrate that LLMs can provide useful formative feedback in programming contexts when guided by structured XAI outputs. These systems not only improve feedback clarity but also support reflective thinking and metacognitive engagement, which are critical for self-regulated learning. Specifically, this approach moves beyond corrective 'feedback' to 'feed-forward' (Lohr et al. 2025), providing students with actionable strategies to regulate their future learning processes.

Nevertheless, there are challenges. As noted by Seo et al. (2025), while LLMs hold great potential in streamlining feedback and assessment, their reliability can fluctuate depending on prompt design and task specificity. To mitigate this, combining XAI's model-based reliability with LLM's communicative power is essential for designing feedback that is both trustworthy and actionable. Future efforts should focus on refining the interface between the two systems to ensure consistency, explainability and educational alignment.

### 2.3 | SRL and AI's Role in Programming Education

SRL refers to a learner's ability to plan, monitor and evaluate their own learning process—skills that are particularly essential in programming education, where students are often required to debug, reflect and iterate independently. Prior research has consistently emphasised the role of SRL in improving programming performance, especially among novice learners who struggle with abstract thinking and problem decomposition (Prasad and Sane 2024).

AI technologies have begun to play an increasingly important role in scaffolding SRL. For instance, behavioural analytics systems can detect disengagement or inefficient strategies in real time and prompt students with corrective feedback (Yang and Ogata 2023). More recently, LLMs have shown promise in enhancing SRL by offering context-aware, timely feedback that encourages reflection, goal setting and adaptation. Sun et al. (2025) provides evidence that GPT-based feedback systems can significantly improve SRL outcomes in computer science courses by supporting students during performance and reflection phases. The emergence of generative AI has also fundamentally shifted the nature of self-regulation. As emphasised by Ng et al. (2024), students now face the challenge of regulating their learning in collaboration with 'AI partners'. This requires a new layer of metacognitive monitoring—not just of one's own cognition, but of the AI's outputs. Our framework addresses this by targeting metacognitive skills like strategy selection (e.g., 'Are you seeking help too often?'), positioning the AI not as a solution-provider, but as a reflective coach. On the other hand, understanding the evolution of these tools is also vital. Before the dominance of commercial copilots, academic-oriented systems were already exploring cloud-based LLMs to assist programming (Tan et al. 2023). Furthermore, generative AI has been shown to reshape pedagogical structures, such as

empowering student-centric questioning in flipped classrooms (Tan 2023). Our study builds on this trajectory by focusing specifically on the metacognitive feedback loop.

However, these tools are not without limitations. While LLMs can offer personalised feedback, the quality and interpretability of that feedback still depend heavily on how well the underlying models are aligned with pedagogical goals. Studies such as Lohr et al. (2025) and Azaiz et al. (2024) note that when LLM-generated feedback is too generic or lacks actionable insight, students may ignore it altogether. Furthermore, without precise grounding data, LLMs risk generating ‘hallucinated’ or overly generic advice that fails to address the specific context of the learner’s problem. Thus, integrating XAI-driven diagnostic insights with LLM-generated suggestions presents a promising path forward—one that combines accurate behavioural analytics with learner-friendly communication. This combination supports deeper SRL by not only identifying what went wrong but also explaining why and how to fix it in a way that aligns with the student’s learning trajectory.

### 3 | Methodology

#### 3.1 | Participants

The participants in the main experiment consisted of 83 undergraduate students enrolled in a general education programming course at a university in Taiwan. Note that this group is distinct from the 516 students in the LBL5-516 dataset that was used to pre-train the predictive model. None of the participants were majoring in science or engineering disciplines. Among them, 39 students were assigned to the experimental group, which received personalised learning analysis reports containing SHAP-based explanations and GPT-generated improvement suggestions throughout the semester. The remaining 44 students were assigned to the control group and did not receive any individualised feedback. We adopted a quasi-experimental design due to the practical constraints of intact classroom settings. Two separate classes were assigned as the experimental group ( $n=39$ ) and the control group ( $n=44$ ). To minimise selection bias, we conducted an independent samples *t*-test on the pre-test scores and found no significant differences in prior programming knowledge ( $p>0.05$ ) or SRL strategies between the two classes. This study was conducted in accordance with the ethical standards of the institutional research committee. Informed consent was obtained from all participants regarding the use of their learning data for research purposes. All personal identifiers were removed and all data were anonymised prior to analysis. While the course was open to all non-STEM majors, the participants were predominantly from the fields of social sciences and humanities. The gender distribution was approximately 60% female and 40% male.

#### 3.2 | Experimental Design

As shown in Figure 1, this study employed a multi-stage experimental design. This process began with Data Preprocessing, where student data was cleaned, encoded and normalised. In the Feature Extraction phase, behavioural and strategy data were categorised into relevant features. Next, during Model Training, classifiers like Random Forest were trained on this data to predict high-risk students. The models were then assessed in the Model Evaluation stage using metrics like accuracy and AUC. For the Explain Generation phase, the model’s predictions were interpreted using XAI (SHAP) and an LLM (GPT-4) to create personalised reports. Finally, in the Explain Evaluation stage, the quality of these explanations was measured using the SCS. The LBL5-516 dataset—comprising students’ learning behaviour and strategy data—was used to train a classification model aimed at identifying high-risk learners. To enhance interpretability, SHAP was applied to identify and explain the most influential features contributing to each prediction. The SHAP outputs were then fed into an LLM, which generated individualised feedback and improvement suggestions for students in the experimental group. These explanations were intended to help learners better understand their learning behaviours and take corrective actions.

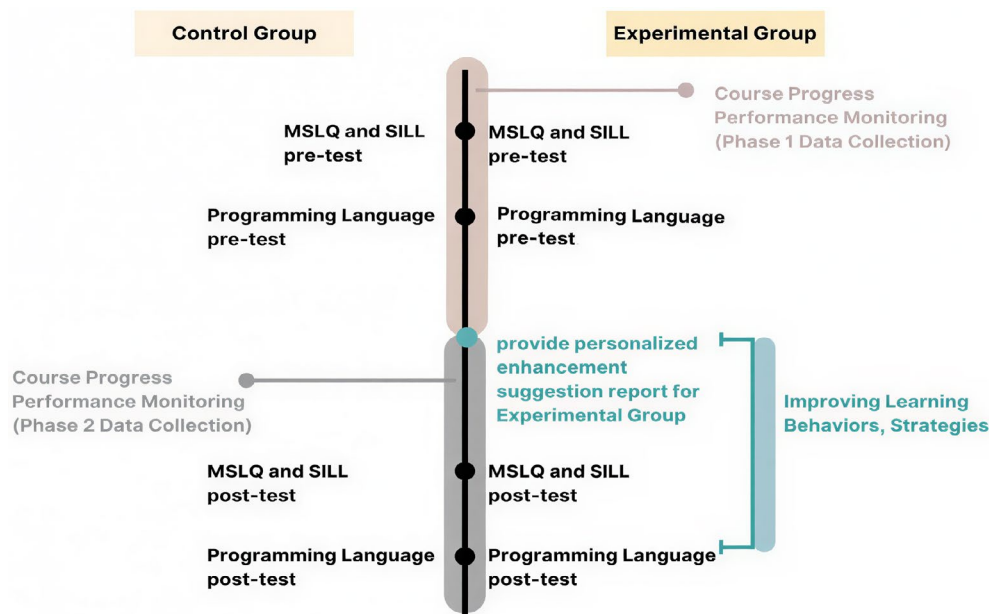
As illustrated in Figure 2, both the experimental and control groups first completed pre-tests using the MSLQ (Motivated Strategies for Learning Questionnaire) and the SILL (Strategy Inventory for Language Learning) to assess students’ learning motivation and strategies at baseline. Throughout the semester, students engaged with two digital platforms: BookRoll, for reading and annotation, and VisCode, a coding environment that logged compilation activity, code errors and usage behaviour. These tools recorded behavioural data in real time, which was used for analysis and feedback generation. Following the mid-term programming assessment, only the experimental group received a personalised learning report. We consider this intervention ‘timely’ as it occurs at the critical midterm pivot point—late enough to utilise robust behavioural data, yet early enough for students to enact strategic changes before the final assessment. The control group, however, continued their coursework as usual without receiving such intervention. Finally, all participants completed the same post-tests (MSLQ, SILL and programming performance test) at the end of the semester to evaluate changes in learning outcomes and self-regulated behaviour.

#### 3.3 | Data Preprocessing

To start, student IDs were encoded to protect privacy. Data cleaning involved removing entries with incomplete learning logs or questionnaire responses (listwise deletion) to ensure data consistency across analyses. Questionnaire responses



**FIGURE 1** | Process of generating personalised improvement suggestions report.



**FIGURE 2** | Experimental procedure of the learning activity.

from MSLQ and SILL were numerically coded and averaged by category (see Table A1 for a full list of strategy features), following the original scale structure. Behavioural logs were collected from BookRoll (e.g., page turns, highlights and notes) and VisCode (e.g., error types, execution attempts and copy-paste) (see Tables A2 and A3 for a full list of behavioural features). All features were retained for completeness (see Appendix A). Next, Min-Max normalisation was applied to scale features between 0 and 1, preventing bias from differing value ranges (Singh and Singh 2020). Finally, students were classified into high-risk (bottom 25%) and low-risk groups based on academic performance (Osmanbegović et al. 2014). Due to strict data privacy protocols and the informed consent agreements signed by participants, the raw dataset and specific processing code cannot be made publicly available to prevent potential re-identification of students. However, to support conceptual replication, detailed descriptions of all extracted features and their definitions are provided in Appendix A (Tables A1–A3).

### 3.4 | Prediction Model and Performance Evaluation

To identify students at risk and support early intervention, we employed three commonly used classification models in educational prediction tasks: Decision Tree (DT), Random Forest (RF) and Support Vector Machine (SVM). These models were selected due to their strong performance and frequent application in prior learning analytics research (López Zambrano et al. 2021; Namoun and Alshantiti 2020). We compared these three models to find which performed best in predicting student risk levels. The most accurate one—whichever it turned out to be—was then used for explanation and generating feedback. Model performance was evaluated using standard classification metrics: accuracy, precision, recall,  $F1$ -score and the Area Under the ROC Curve (AUC) to assess predictive accuracy and discriminatory ability. To ensure the robustness of the model and minimise

overfitting, we employed a 5-fold cross-validation technique during the training phase. Hyperparameters for the classifiers (e.g., number of trees and maximum depth for Random Forest) were optimised using a grid search approach to identify the most effective configuration for our dataset.

The purpose of this classification was not to label students, but to create a predictive task that would allow the XAI model to identify which learning behaviours were most influential for a given outcome. These influential features, identified by SHAP, were then explained to all students, regardless of their risk category, to foster self-regulation. Note that the ‘high-risk’ group was defined as students in the bottom 25% based on their academic performance.

### 3.5 | Generation of Explanations and Improvement Suggestions

To better interpret the predictions produced by our model, we applied SHapley Additive exPlanations (SHAP), a widely used method that attributes feature importance based on cooperative game theory. By evaluating each possible feature combination, SHAP offers a consistent way to explain how much each input contributed to the final prediction (Lundberg and Lee 2017). One of the reasons we chose SHAP is because it provides both global-level summaries and local explanations. Summary plots help instructors see overall patterns, while waterfall plots visualise how individual features influenced each student’s predicted risk level. Since every learner’s behaviour is different, these personalised plots can help students see, more clearly, what aspects of their study habits are helping—or hurting—their outcomes.

Following the framework proposed by Hsu et al. (2024), we implemented this integration as a ‘two-stage pipeline’. In the first stage, SHAP calculates the precise numerical contribution of each feature. In the second stage, these values are

used as structured contextual prompts for the large language model (GPT-4) to turn these SHAP explanations into plain-language feedback and improvement suggestions. By feeding the key feature contributions into the model, GPT-4 generated text that explained not just what features mattered, but also why they mattered—bridging the gap between raw data and learner understanding. These SHAP outputs were then processed by GPT-4, which leverages a transformer architecture to generate natural language feedback. This approach combines SHAP's diagnostic precision with GPT's adaptability to create actionable feedback for learners. The feedback generated by GPT-4 was personalised; it specifically highlighted certain learning strategies that needed improvement based on the SHAP analysis. For the purpose of analysis, we define:

- *Guided Features*: The specific strategy dimensions (e.g., 'Help-seeking') that were explicitly identified and addressed in the student's personalised report.
- *Unguided Features*: The strategy dimensions that were not mentioned in the report.

### 3.6 | Evaluation of the Quality of Explanations

To assess the perceived quality of explanations generated by XAI and LLM systems, we adopted the System Causability Scale (SCS). The SCS evaluates how well users can understand and also trust the causal reasoning provided in AI-driven feedback. A total of 39 participants completed the questionnaire (Table 1). Developed by Holzinger et al. (2019, 2020), the SCS extends traditional usability models—like the System Usability Scale—by adding a focus on causability, which emphasises whether

explanations can support human reasoning and informed decision-making.

To address RQ1 regarding the interpretability of different feedback formats, the 39 students in the experimental group were randomly assigned to two conditions for the survey evaluation only: *Visual Explanation Group* ( $n=20$ ) that received the SHAP-based waterfall plots showing how different behaviour features contributed to their learning predictions. *Textual Explanation Group* ( $n=19$ ), in contrast, received GPT-4-generated text explanations based on the same SHAP input. The subset of students ( $n=22$ ) who completed both pre- and post-tests were later included in the learning outcome analysis (RQ2 and RQ3). Regardless of type, all students received improvement suggestions generated by GPT-4. Before rating the explanations, students were briefly instructed on how to read SHAP plots and interpret LLM-generated content—though in practice, some probably still felt more comfortable with one format than the other, which is hard to avoid.

## 4 | Results and Discussion

### 4.1 | Model Performance Examination

In this study, we compared three common predictive models—DT, RF and SVM—to forecast whether students were at high risk based on their learning behaviours and strategies. The models were trained separately on the two types of data, and their performance was evaluated using standard classification metrics. Table 2 shows the performance results across both datasets.

As shown in Table 2, Random Forest (RF) demonstrated the most balanced and superior performance on both the learning behaviour data ( $AUC=0.709$ ) and the learning strategy data ( $AUC=0.874$ ). It consistently outperformed Decision Tree (DT) and Support Vector Machine (SVM) across most metrics, aligning with prior research suggesting the robustness of ensemble models in educational prediction. Therefore, RF was selected for the explanation generation phase. It is important to note that this comparison of predictive accuracy was a preliminary technical step to ensure the reliability of the underlying model; it does not address the RQ 1, which focuses on the interpretability of the feedback methods (discussed in Section 4.3). These results align with findings in previous studies, where ensemble models like RF consistently outperform simpler classifiers in educational prediction tasks due to their robustness and noise tolerance (Tomasevic et al. 2020).

Overall, the results show that RF tends to produce more reliable predictions in both learning behaviour and strategy datasets. While SVM also demonstrated strong performance, especially in precision, its margin-based approach may be more sensitive to the specific feature distribution, and DT's simplicity perhaps limits its predictive power in complex learning data scenarios.

### 4.2 | Explanations Generated by SHAP

To further explore individual learning conditions, SHAP waterfall plots were used to visualise the contribution of each

**TABLE 1** | The question list for SCS.

| Question list                                                                                                  |
|----------------------------------------------------------------------------------------------------------------|
| 1. I found that the data included all relevant known causal factors with sufficient precision and granularity. |
| 2. I understood the explanations within the context of my work.                                                |
| 3. I could change the level of detail on demand.                                                               |
| 4. I did not need support to understand the explanations.                                                      |
| 5. I found the explanations helped me to understand causality.                                                 |
| 6. I was able to use the explanations with my knowledge base.                                                  |
| 7. I did not find inconsistencies between explanations.                                                        |
| 8. I think that most people would learn to understand the explanations very quickly.                           |
| 9. I did not need more references in the explanations: e.g., medical guidelines, and regulations.              |
| 10. I received the explanations in a timely and efficient manner                                               |
| $SCS = \sum_i \text{Rating}_i / 50$                                                                            |

behavioural and strategic feature to the model's prediction. As shown in Figure 3, the plot highlights the impact of different features on one student's predicted learning risk. On the left, each feature and its corresponding value are displayed. The x-axis represents the log-odds (or probability contribution) of the model's prediction. Negative values (blue bars) indicate that the feature value contributed to lowering the predicted risk (i.e., a protective factor), whereas the Positive values (red bars) indicate that the feature contributed to increasing the predicted risk. While a SHAP value of 0.03 appears small numerically, in the context of our model's probability output, it represents a meaningful contribution to the risk classification relative to other features. The numerical value above each arrow reflects the magnitude of that contribution.

This approach follows the growing practice of using local explanations to support actionable feedback in learning analytics (Lundberg and Lee 2017; Kay 2023). In this case, SHAP allowed us to identify 'affective strategy' as a critical factor influencing the prediction outcome, with a feature value of 0.222 and a SHAP value of 0.03—indicating a modest but still notable positive impact on high-risk labelling.

Following this, we leveraged an LLM (GPT-4) to generate natural language explanations based on the SHAP outputs. The

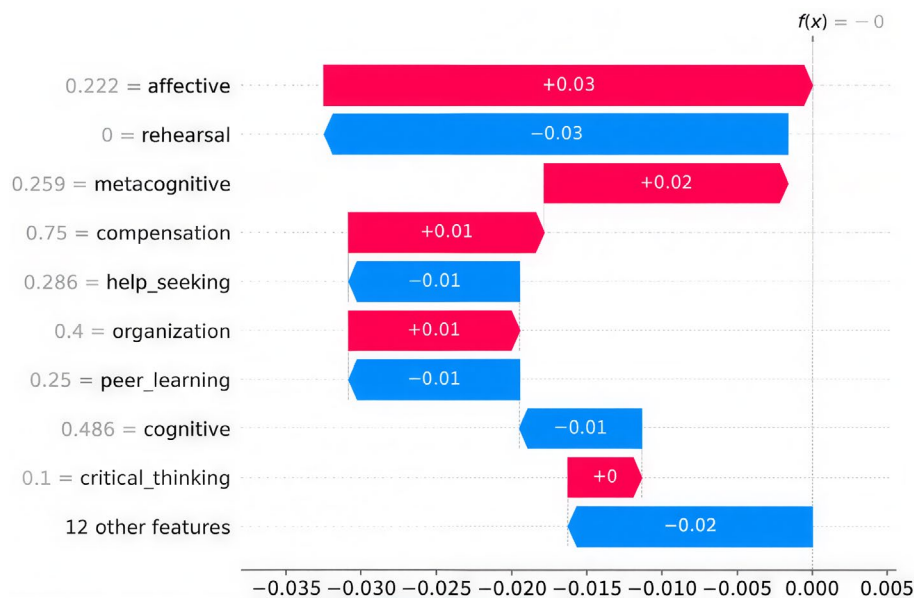
left portion of the output includes a breakdown of key influencing features and why they mattered; the right provides personalised suggestions for behavioural and strategic adjustments. As shown in Figure 4, the LLM-generated explanation for this specific student emphasised emotional regulation and motivation strategies, offering insights into how low use of affective strategies might increase academic risk. It is important to note that this is an illustrative example; the actual feedback generated for each student varied dynamically based on their specific high-risk features (e.g., a student with low code compilation frequency would receive specific strategies on coding practice rather than emotional regulation). While not every suggestion may be perfect, the LLM demonstrated the ability to contextualise model logic and suggest possible learning improvements, as supported in recent XAI and education research (Holzinger et al. 2020).

### 4.3 | Exploring Interpretability and Perceived Causality

To address RQ1 (i.e., whether LLMs offer superior interpretability compared to XAI in programming education), this study employed both SHAP (as a representative XAI technique) and GPT-4 (as an LLM) to provide explanatory feedback to students.

**TABLE 2** | Predictive results of identifying high-risk students.

|                    | Method | Accuracy | Precision | Recall | F1    | AUC   |
|--------------------|--------|----------|-----------|--------|-------|-------|
| Learning behaviour | DT     | 0.678    | 0.661     | 0.678  | 0.669 | 0.525 |
|                    | RF     | 0.833    | 0.822     | 0.833  | 0.821 | 0.709 |
|                    | SVM    | 0.833    | 0.840     | 0.833  | 0.803 | 0.659 |
| Learning strategy  | DT     | 0.884    | 0.917     | 0.884  | 0.895 | 0.847 |
|                    | RF     | 0.930    | 0.937     | 0.930  | 0.933 | 0.874 |
|                    | SVM    | 0.860    | 0.779     | 0.860  | 0.817 | 0.687 |



**FIGURE 3** | Waterfall plot of the students' learning strategies.

|   | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Enhancement suggestion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <p><b>Affective strategy:</b><br/>With a feature value of 0.222 and a SHAP value of 0.03, this indicates lower performance of this feature, thus increasing learning risk. Possible reasons may include:</p> <p>1. Difficulty in regulating emotions: If students lack affective strategies, they may be more susceptible to negative emotions such as anxiety, frustration, or encouragement. These negative emotions can disrupt their learning process, reducing their motivation and focus.</p> <p>2. Lack of motivation and interest: If students lack affective strategies, they may perceive learning as less engaging, leading to decreased enthusiasm for learning.</p> <p>If the student can improve the use of their affective strategies, it may potentially lower the risk of them becoming a high-risk student.</p> | <p><b>Embracing a challenging mindset:</b> Firstly, students should approach challenges with a positive attitude, recognizing that encountering problems is part of the learning process rather than a symbol of failure. They can remind themselves that challenges can help them grow and progress.</p> <p><b>Emotional self-regulation:</b> Students may feel frustrated, anxious, or discouraged when facing problems, but they can learn to regulate their emotions. This includes deep breathing, relaxation techniques, or adopting a positive mindset to stay calm and focused.</p> <p><b>Seeking support:</b> Students should not hesitate to seek help; they can reach out to teachers, classmates, or online forums for support and advice. Sharing problems and difficulties with others can make them feel less isolated while gaining new perspectives and solutions.</p> <p><b>Conclusion:</b> When students encounter challenges in Python courses, they can employ emotional strategies such as embracing challenges, regulating emotions, seeking support, breaking down problems, and persevering to overcome difficulties and enhance learning outcomes.</p> |

**FIGURE 4** | Example of LLM-based explanation and enhancement suggestions.

We assessed the quality of these explanations using the SCS, which evaluates how well users understand the causal reasoning behind AI-generated feedback (Holzinger et al. 2020).

The results are presented in Table 3. Participants in the Textual Explanation Group (LLM), who received GPT-4-generated explanations, consistently scored higher across several SCS dimensions compared to the Visual Explanation Group (XAI). Specifically, for the learning behaviour explanations ( $n = 39$ ), the textual format led to significantly better understanding of data factors and knowledge integration. Similarly, for the subset of students with learning strategy data ( $n = 22$ ), the textual explanations outperformed the visual SHAP plots in terms of facilitating rapid comprehension and understanding causality. It is worth noting that the Textual group scored slightly lower on 'Effectiveness' compared to 'Detail Level'. This likely reflects the cognitive load required to read detailed text compared to glancing at a visual plot; however, the higher scores in comprehension suggest that this trade-off is valuable for deep learning. These findings align with Rawal et al. (2023), who argue that LLMs, when guided by structured feature inputs, can enhance learners' causal understanding more effectively than standalone XAI.

This finding aligns with recent studies by Hsu et al. (2024), which demonstrate how SHAP outputs can be transformed into pedagogically meaningful explanations using LLMs, thereby enhancing transparency and guiding student learning. Additionally, as Seo et al. (2025) emphasises, XAI systems should be designed to support user comprehension and trust, especially in educational environments where clarity of explanation influences student engagement and learning outcomes. As highlighted by Barman et al. (2024), while LLMs offer new forms of explanation and

**TABLE 3** | SCS rating for XAI and LLM.

| Dimensions                               | Learning behaviour explanations (N = 39) |               | Learning strategy explanations (N = 22) |               |
|------------------------------------------|------------------------------------------|---------------|-----------------------------------------|---------------|
|                                          | Visual (XAI)                             | Textual (LLM) | Visual (XAI)                            | Textual (LLM) |
| 1. Data factors                          | 3.67                                     | 4.14          | 3.69                                    | 3.89          |
| 2. Understanding                         | 3.6                                      | 4.21          | 3.69                                    | 4.44          |
| 3. Detail level                          | 3.67                                     | 4.14          | 4                                       | 3.89          |
| 4. No support needed                     | 3.07                                     | 4.07          | 3.08                                    | 3.67          |
| 5. Understanding causality               | 3.73                                     | 4.29          | 3.85                                    | 4.33          |
| 6. Knowledge integration                 | 3.4                                      | 4             | 3.77                                    | 3.89          |
| 7. No inconsistencies                    | 3.4                                      | 3.93          | 3.38                                    | 3.67          |
| 8. Rapid understanding                   | 3.2                                      | 4.07          | 3.46                                    | 4.11          |
| 9. No reference needed                   | 2.93                                     | 3.93          | 3.15                                    | 4.11          |
| 10. Effectiveness                        | 3.67                                     | 4.36          | 3.54                                    | 3.56          |
| SCS Score: $\sum_i \text{Rating}_i / 50$ | 0.687                                    | 0.823         | 0.712                                   | 0.791         |

feedback, they also risk generating overconfident or misleading outputs if not supported by adequate user guidance and contextualisation. In educational settings, this may hinder rather than help learners if outputs are not sufficiently interpretable or aligned with instructional goals. Thus, our study reinforces the importance of combining XAI's reliable diagnostic outputs with LLM's linguistic expressiveness to balance interpretability and instructional value in AI-based education systems.

#### 4.4 | Assessing the Impact of LLM-Generated Improvement Suggestions on Self-Regulation and Programming Language Learning Strategies

To address RQ2 (i.e., can LLM-generated improvement suggestions enhance students' self-regulation and programming language learning strategies?), we compared pre- and post-test scores from the MSLQ and SILL questionnaires. Of the participants, 22 students had complete pre- and post-test data. Among these, each student received personalised improvement suggestions generated by GPT-4, based on SHAP analysis of their learning behaviours and strategies. Note that we analysed the improvement in SRL scores at the feature level. We compared the mean improvement (post-test minus pre-test) of the *Guided Features* against the *Unguided Features*. A paired sample *t*-test was conducted to determine if specific algorithmic guidance led to greater strategy adoption compared to non-targeted dimensions.

The results shown in Table 4 indicate a significant difference in learning gains between features that were guided by LLM-generated suggestions (*Guided Features*) and those that were not (*Unguided Features*). Specifically, guided features improved by an average of 0.566, compared to 0.214 for unguided features ( $t = 5.903$ ,  $p < 0.001$ ). This significant improvement provides empirical evidence that the personalised text feedback directly influenced students' subsequent learning behaviours, moving beyond mere score reporting to actual behavioural modification. This also suggests that targeted, AI-generated feedback plays a meaningful role in supporting SRL strategy development. Sun et al. (2025) similarly observed that LLM-driven formative feedback improved students' ability to adapt learning behaviours and metacognitive strategies.

Table 5 presents comparisons across specific strategy dimensions. Statistically significant improvements were observed in dimensions such as intrinsic motivation, control beliefs, rehearsal, critical thinking, help-seeking, memory and compensation. These results align with Lohr et al. (2025) and Azaiz et al. (2024), who found that LLMs can deliver actionable and personalised advice that effectively supports learning habits in programming contexts.

**TABLE 4** | Differences in the average between pre-test and post-test of learning strategy scores.

| Group    | N  | Mean  | Std   | <i>t</i> | <i>p</i> |
|----------|----|-------|-------|----------|----------|
| Guided   | 22 | 0.566 | 0.255 | 5.903    | <0.001   |
| Unguided | 22 | 0.214 | 0.117 |          |          |

The benefits were particularly evident in behaviourally actionable dimensions (e.g., rehearsal and help-seeking), likely because these are more susceptible to concrete guidance. In contrast, motivational or affective constructs like test anxiety showed limited change, indicating the need for longer-term or multimodal interventions. Overall, these findings reinforce the potential of integrating LLMs into SRL support systems, provided they are designed to align feedback with specific cognitive and behavioural targets.

#### 4.5 | Evaluating the Effects of LLM-Generated Improvement Suggestions

To address the RQ3 (Can LLM-generated improvement suggestions improve students' learning performance in programming education?), we compared pre- and post-test programming scores between students who received LLM-generated suggestions (experimental group) and those who did not (control group). While pre-test scores showed no significant difference (Mean = 63.90 vs. 64.97,  $p > 0.05$ ), post-test results revealed a large gap: the experimental group scored significantly higher (Mean = 87.53) than the control group (Mean = 68.42), with  $t = -5.946$ ,  $p < 0.001$ .

These results shown in Table 6 indicate that personalised improvement suggestions generated by GPT-4 had a meaningful effect on students' programming achievement. This aligns with the findings of Nguyen and Allan (2024), who demonstrated that GPT-4-based formative feedback could lead to measurable performance gains in Java programming tasks. Furthermore, Koutchme et al. (2025) emphasise that the integration of LLMs for feedback generation enhances clarity and reduces cognitive load in programming assignments, which may explain why learners in the experimental group were able to internalise suggestions more effectively and perform better in assessments. The substantial performance gap suggests that the intervention did more than provide answers; by specifically targeting high-impact regulatory strategies (e.g., Help-Seeking, Rehearsal) identified by the SHAP model, the feedback unlocked 'bottle-necks' in student learning processes that generic feedback typically misses.

Taken together, the findings suggest that well-designed LLM-enhanced feedback systems can significantly improve programming learning outcomes. However, consistent with Jia et al. (2024), it remains crucial to monitor how students interpret and apply AI-generated feedback to ensure long-term efficacy and avoid misapplication. Additional studies with varied populations and programming domains are recommended to validate and generalise these results. Furthermore, a comparative analysis highlights the resource efficiency of this approach. Providing detailed, manual feedback for 83 students would typically require an estimated 25–30 h of instructor time (approximately 20 min per student), often leading to delays. In contrast, our automated framework generated personalised reports for the entire cohort in minutes. While human feedback remains the 'gold standard' for depth, the immediacy and scalability of the AI system ensure that every student receives timely support—a critical factor for effective SRL interventions.

**TABLE 5** | Effects on learning strategy dimensions: A comparison of guided and unguided features.

| Questionnaire           | Feature                       | Group    | N  | Mean  | Std   | t      | p      |
|-------------------------|-------------------------------|----------|----|-------|-------|--------|--------|
| SRL learning motivation | intrinsic                     | Guided   | 6  | 0.755 | 0.541 | 3.144  | 0.024  |
|                         |                               | Unguided | 16 | 0.082 | 0.136 |        |        |
|                         | extrinsic                     | Guided   | 9  | 0.028 | 0.083 | 0.262  | 0.796  |
|                         |                               | Unguided | 13 | 0.019 | 0.069 |        |        |
|                         | control_beliefs               | Guided   | 11 | 0.379 | 0.187 | 4.407  | <0.001 |
|                         |                               | Unguided | 11 | 0.073 | 0.134 |        |        |
|                         | self_efficacy                 | Guided   | 5  | 0.075 | 0.168 | 0.940  | 0.358  |
|                         |                               | Unguided | 17 | 0.022 | 0.091 |        |        |
| SRL learning strategy   | test_anxiety                  | Guided   | 6  | 0.233 | 0.480 | 1.124  | 0.312  |
|                         |                               | Unguided | 16 | 0.013 | 0.050 |        |        |
|                         | rehearsal                     | Guided   | 13 | 0.279 | 0.240 | 3.276  | 0.005  |
|                         |                               | Unguided | 9  | 0.046 | 0.073 |        |        |
|                         | elaboration                   | Guided   | 3  | 0.222 | 0.385 | 0.792  | 0.508  |
|                         |                               | Unguided | 19 | 0.044 | 0.156 |        |        |
|                         | critical_thinking             | Guided   | 5  | 0.245 | 0.247 | 2.282  | 0.034  |
|                         |                               | Unguided | 17 | 0.068 | 0.118 |        |        |
|                         | metacognitive_self_regulation | Guided   | 9  | 0.222 | 0.118 | 0.702  | 0.491  |
|                         |                               | Unguided | 13 | 0.167 | 0.215 |        |        |
|                         | time_and_study_environment    | Guided   | 8  | 0.406 | 0.265 | −0.856 | 0.402  |
|                         |                               | Unguided | 14 | 0.554 | 0.440 |        |        |
|                         | effort_regulation             | Guided   | 4  | 0.875 | 0.478 | 0.130  | 0.898  |
|                         |                               | Unguided | 18 | 0.833 | 0.594 |        |        |
|                         | peer_learning                 | Guided   | 9  | 0.083 | 0.132 | 0.382  | 0.707  |
|                         |                               | Unguided | 13 | 0.060 | 0.144 |        |        |
| SILL                    | help_seeking                  | Guided   | 17 | 0.662 | 0.232 | 3.11   | 0.005  |
|                         |                               | Unguided | 5  | 0.300 | 0.209 |        |        |
|                         | memory                        | Guided   | 11 | 0.844 | 0.542 | 3.346  | 0.005  |
|                         |                               | Unguided | 11 | 0.255 | 0.217 |        |        |
|                         | cognitive                     | Guided   | 8  | 0.760 | 0.492 | 0.131  | 0.897  |
|                         |                               | Unguided | 14 | 0.731 | 0.500 |        |        |
|                         | compensation                  | Guided   | 11 | 0.818 | 0.593 | 2.459  | 0.025  |
|                         |                               | Unguided | 11 | 0.301 | 0.364 |        |        |
|                         | metacognitive                 | Guided   | 18 | 1.204 | 0.246 | 0.399  | 0.694  |
|                         |                               | Unguided | 4  | 1.028 | 0.858 |        |        |
|                         | affective                     | Guided   | 19 | 0.890 | 0.492 | 1.826  | 0.083  |
|                         |                               | Unguided | 3  | 0.347 | 0.334 |        |        |
|                         | social                        | Guided   | 18 | 0.745 | 0.478 | 1.308  | 0.206  |
|                         |                               | Unguided | 4  | 0.417 | 0.289 |        |        |

**TABLE 6** | The results of the pre-test and post-test scores between experimental and control group.

| Test      | Group        | N  | Mean   | Std    | t      | p       |
|-----------|--------------|----|--------|--------|--------|---------|
| Pre-test  | Control      | 44 | 63.898 | 14.238 | −0.327 | 0.745   |
|           | Experimental | 39 | 64.974 | 15.794 |        |         |
| Post-test | Control      | 44 | 68.423 | 17.892 | −5.946 | < 0.001 |
|           | Experimental | 39 | 87.526 | 10.898 |        |         |

## 5 | Conclusions

This study explored the integration of LLMs and XAI to provide personalised feedback in programming education. By combining SHAP-based interpretability with GPT-4-generated suggestions, we examined how AI-driven explanations could support learning behaviours, strategies and performance. Results showed that LLM-generated feedback received higher SCS scores than traditional XAI visualisations, suggesting stronger clarity and perceived usefulness (Holzinger et al. 2020). Students who received LLM-based suggestions significantly improved their self-regulation strategies and post-test programming scores, showing the potential of adaptive, explanation-driven feedback to facilitate self-regulated learning processes. Rather than simply combining tools, the results suggest that integrating LLMs enhances the interpretability of XAI, making evidence-based feedback more accessible and actionable for learners. These findings also echo previous arguments that causability may be more important than model transparency in educational contexts (Kay 2023).

The results of this research contribute to the growing body of evidence supporting AI-supported personalised learning, particularly in technical subjects like programming where individualised guidance can be difficult to scale. The use of LLMs not only addressed the interpretability gap found in many visual-based XAI tools but also enhanced students' ability to act on feedback, especially in SRL dimensions such as help-seeking, rehearsal and metacognitive monitoring. This aligns with recent studies suggesting that LLM-generated explanations can bridge cognitive and technical barriers in digital learning environments (Sun et al. 2025; Jia et al. 2024). To address concerns regarding feedback accuracy, we propose a 'Teacher-in-the-Loop' oversight model for future deployment. Rather than fully automating the process, a dashboard could flag 'low-confidence' reports or 'high-risk' student cases for instructor review. This allows educators to act as efficient gatekeepers—vetting or customising the AI's suggestions—thereby balancing the scalability of LLMs with the pedagogical reliability of human expertise.

Still, the study has its limitations. The short-term design and limited sample size restrict generalisability and statistical power. The participants were non-STEM majors, which may influence how results would transfer to computer science students or professional learners. As noted by Youn et al. (2025), the effectiveness of AI-generated feedback may differ across settings and over time. Future research should explore long-term impacts, learner trust development and broader applications of LLM-XAI integration, including adaptive systems that fine-tune feedback based on learner progress, goals and motivation

profiles. Theoretically, this feedback loop aligns with principles of Reinforcement Learning with Human Feedback. As discussed by Wong and Tan (2024), just as crowd-sourced feedback aligns LLMs with human intent, student behavioural responses (e.g., adopting a suggested strategy) can serve as a 'reward signal'. Future systems could use this data to fine-tune the LLM, ensuring that generated advice becomes increasingly relevant and effective for promoting self-regulation.

### Author Contributions

C.C.Y.Y. and M.L. designed and carried out the research studies. A.Y.Q.H. participated in the conduction and discussions related to data collection and analysis. C.C.Y.Y. drafted the manuscript. M.L. and A.Y.Q.H. supervised this research and contributed to the review and discussion of the manuscript. All authors read and approved the final manuscript.

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During the preparation of this work, the author(s) used ChatGPT in order to improve overall language and readability, with caution. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

### Conflicts of Interest

The authors declare no conflicts of interest.

### Data Availability Statement

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

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## Appendix A

**TABLE A1** | Learning strategies features extracted from learning questionnaires.

| Questionnaire           | Feature                       | Description                                                                                                                    |
|-------------------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| SRL learning motivation | intrinsic                     | The internal drive and enjoyment individuals experience when engaging in learning activities.                                  |
|                         | extrinsic                     | External factors, such as rewards or recognition, influence individuals' engagement in learning activities.                    |
|                         | task_value                    | Assess students' perceptions of the interest, importance and usefulness of course content.                                     |
|                         | control_beliefs               | Assesses whether students believe their hard work will lead to positive outcomes.                                              |
|                         | self_efficacy                 | Assesses the judgement and confidence that the student can complete the task independently.                                    |
|                         | test_anxiety                  | Student anxiety levels about tests in the course.                                                                              |
| SRL learning strategy   | rehearsal                     | Repetitive or repeated review of study material to enhance retention and understanding.                                        |
|                         | elaboration                   | Enhancing understanding by making connections and creating meaningful associations with prior knowledge.                       |
|                         | organisation                  | The act of structuring and arranging information in a systematic and logical manner to facilitate comprehension and retrieval. |
|                         | critical_thinking             | The process of objectively analysing and evaluating information to make informed judgements and decisions.                     |
|                         | metacognitive_self_regulation | The skill of monitoring and controlling one's own learning process for better outcomes.                                        |
|                         | time_and_study_environment    | Encompass managing study time effectively and creating an optimal setting for focused learning.                                |
|                         | effort_regulation             | The skill of consciously managing and adjusting one's level of effort to maximise learning outcomes.                           |
|                         | peer_learning                 | The process of students learning from and with their peers, through collaborative activities and discussions.                  |
| SILL                    | help_seeking                  | Students seek support from others when they encounter difficulties in their studies.                                           |
|                         | memory                        | The ability and strategies used to effectively remember and recall the materials learned.                                      |
|                         | cognitive                     | The mental processes and abilities involved in learning, such as attention, memory, thinking and problem-solving.              |
|                         | compensation                  | The use of alternative strategies or resources to overcome difficulties or limitations in language skills or knowledge.        |
|                         | metacognitive                 | Students will plan, organise, evaluate and monitor their own language learning.                                                |
|                         | affective                     | Students regulate their emotions, motivations and attitudes when learning a language.                                          |
|                         | social                        | Student interacts with others while learning.                                                                                  |

**TABLE A2** | Learning behaviours features extracted from BookRoll.

| Feature         | Description                                                                  |
|-----------------|------------------------------------------------------------------------------|
| ADD BOOKMARK    | Added a bookmark to current page.                                            |
| ADD MARKER      | Added a marker to current page.                                              |
| ADD MEMO        | Added a memo to current page.                                                |
| ADD_HW_MEMO     | Added a handwrite memo to current page.                                      |
| BOOKMARK_JUMP   | Jump to a specific page with a bookmark.                                     |
| CHANGE MEMO     | Modify the content of an existing memo on current page.                      |
| CLEAR_HW_MEMO   | Clear the content of an existing handwrite memo on current page.             |
| CLOSE           | Closed the book.                                                             |
| DELETE BOOKMARK | Deleted a bookmark on current page.                                          |
| DELETE MARKER   | Deleted a marker on current page.                                            |
| DELETE_MEMO     | Deleted a memo on current page.                                              |
| GETIT           | Press the smiley face icon to indicate the understanding on current page.    |
| MEMO_JUMP       | Select a note to jump to the specific page.                                  |
| NEXT            | Went to the next page.                                                       |
| NOTGETIT        | Press the crying face icon to indicate the misunderstanding on current page. |
| OPEN            | Opened the book.                                                             |
| PAGE_JUMP       | Jumped to a particular page.                                                 |
| PREV            | Went to the previous page.                                                   |
| SEARCH          | Searched for something within the e-book.                                    |
| SEARCH_JUMP     | Jumped to a page from the search results.                                    |
| UNDO_HW_MEMO    | Undo the last action of handwriting.                                         |

**TABLE A3** | Learning behaviours features extracted from VisCode.

| Feature        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| code_length    | Number of lines of code (LOC) in this semester.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| code_copy      | A number of times a student copies codes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| code_execution | Number of times a student executes codes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| code_paste     | A number of times a student paste codes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| code_speed     | Average input digits per minute.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| notebook_open  | A number of times a student opens a coding environment.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| tree_open      | A number of times a student opens a folder looking for a code.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Error          | AssertionError, AttributeError, AxisError, ConversionError, DataError, EmptyDataError, Exception, FileExistsError, FileNotFoundError, HTTPError, ImportError, IndentationError, IndexError, IndexingError, InvalidURL, JSONDecodeError, KeyError, KeyboardInterrupt, LookupError, ModuleNotFoundError, NameError, OperationalError, OSError, OverflowError, ParserError, RecursionError, RuntimeError, SelectorSyntaxError, SpecificationError, StopIteration, SyntaxError, TabError, TclError, TypeError, UFuncTypeError, UndefinedVariableError, UnboundLocalError, UnicodeDecodeError, ValueError, WebDriverException, XLRError, ZeroDivisionError. |